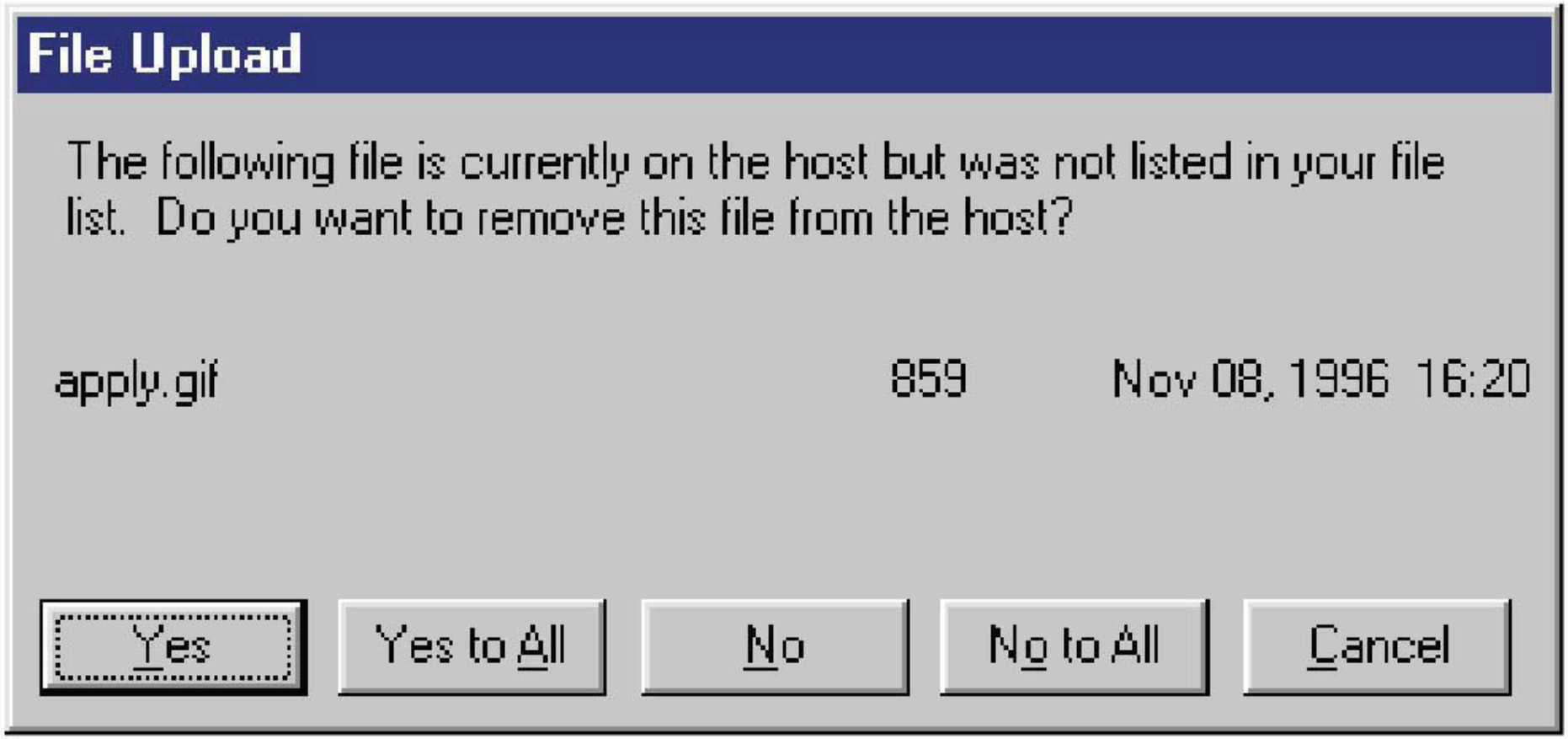
Efficiency

WARNING
Cookie:rambo amadeus@microsoft.com is a Cookie!! Are you sure you want to delete it?
Yes
No

WARNING
Cookie:rambo amadeus@microsoft.com is a Cookie!! Are you sure you want to delete it?
Yes
No

Suppose you want to delete multiple cookie files (older Internet Explorer)



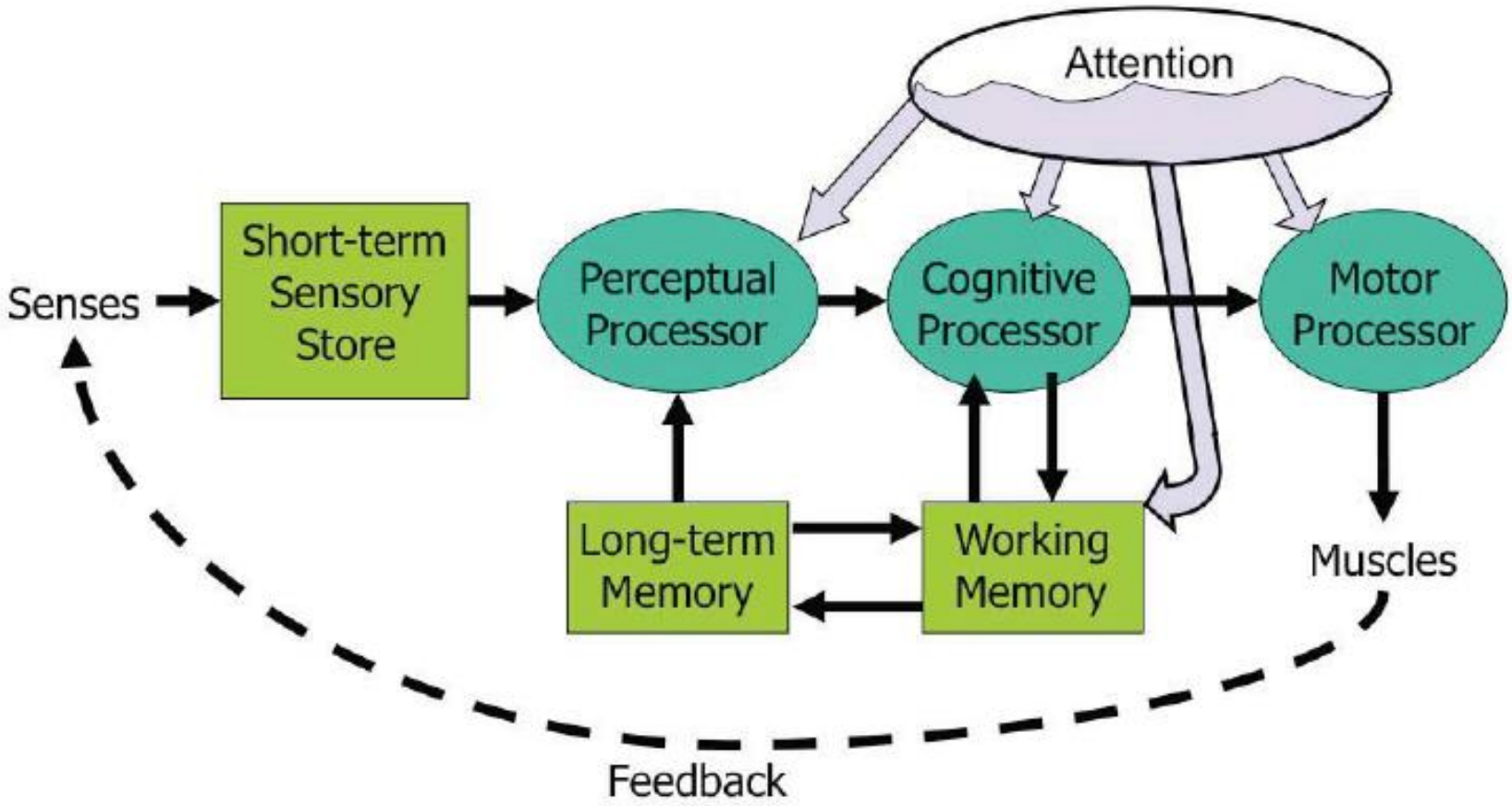
what if you know there's a file on the host that you **don't** want to delete?

A nice solution offered in Eclipse

Today's Topics

- Human information processing
- Pointing efficiency
  - Fitts's Law & Steering Law
- Design principles
  - Shortcuts
  - Defaults, history, and anticipation
  - Anticipation
- Predicting efficiency
  - Keystroke-level model

Human Information Processing



Perceptual (algısal) processor takes the stored sensory input and tries to recognize *symbols*: letters, words, phonemes, icons. It is aided by the **long-term memory**

cognitive (bilişsel) processor takes the symbols recognized by the perceptual processor and makes comparisons and decisions

Processors

- Processors have a cycle time
  - T_p ~ 100ms [50-200 ms]
  - T_c ~ 70ms [30-100 ms]
  - T_m ~ 70ms [25-170 ms]

- Processor speed varies by person and conditions
  - Fastest may be 10x slowest

time needed to accept one input and produce one output

Perceptual Fusion

- Two stimuli within the same PP cycle (T_p ~ 100ms) appear **fused**
  - Causality is strongly influenced by fusion

Cognitive processor

Types of decision making:

rule-based, in which the human is consciously processing a set of rules of the form *if X, then do Y* (a novice driver thinking of what to do when approaching an intersection…)

skill-based decision making: A skill is a procedure that has been learned thoroughly from practice; walking, talking, pointing, reading, driving, typing

knowledge-based, which involves much higher-level thinking and problem-solving. (figuring out why your car won’t start…)

Motor Processing

- Open-loop control
  - Motor processor runs a program by itself
  - cycle time is $T_m \sim 70$ ms
- Closed-loop control
  - Muscle movements (or their effect on the world) are perceived and compared with desired result
  - cycle time is $T_p + T_c + T_m \sim 240$ ms

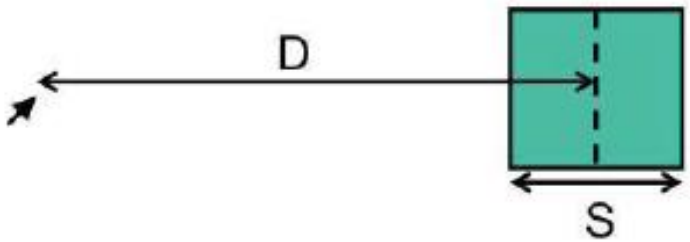
Example of closed-loop full processing cycle

choice – choosing a different response for each stimulus

Fitts's Law

- Fitt's Law
  - Time T to move your hand to a target of size S at distance D away is:

$$T = RT + MT = a + b \log (D/S + 1)$$

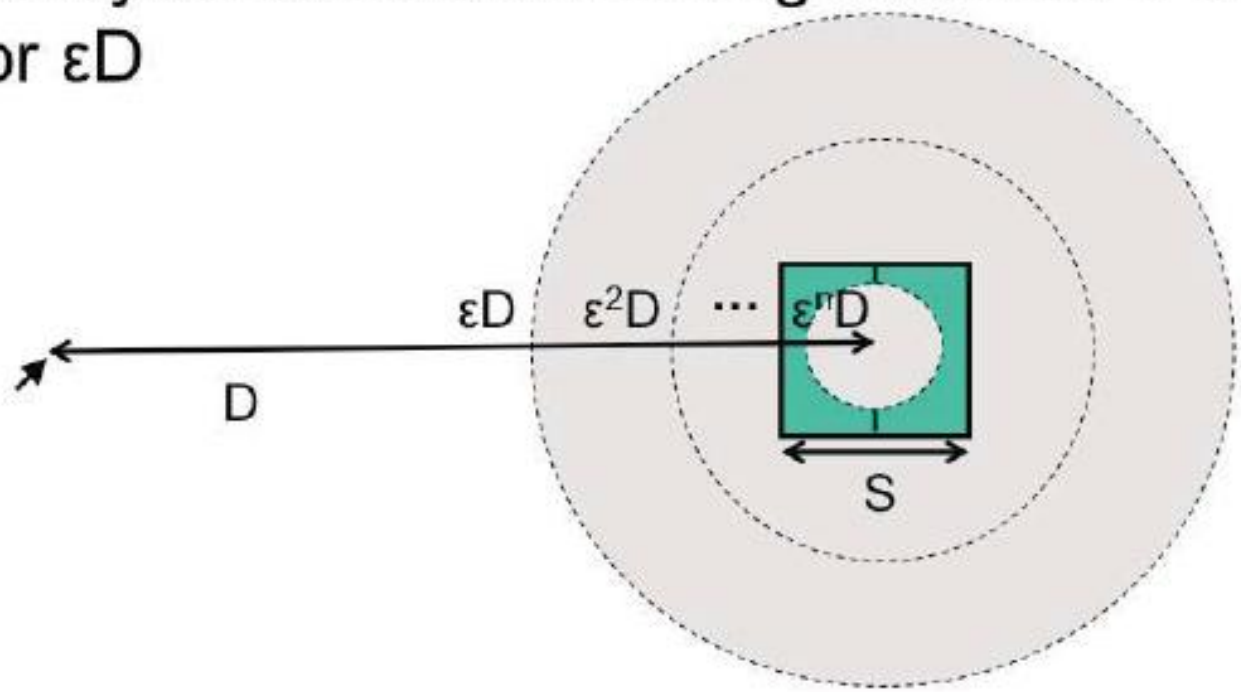


  - Depends only on *index of difficulty* log(D/S + 1)

applies equally well to using a mouse to point at a target on a screen (react + movement)

Explanation of Fitts's Law

- Moving your hand to a target is closed-loop control
- Each cycle covers remaining distance D with error εD

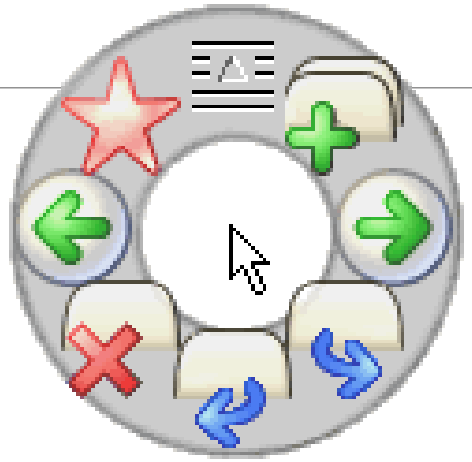


Implications of Fitts's Law

- Targets at screen edge are easy to hit
  - Mac menubar beats Windows menubar
  - Unclickable margins are foolish
- Linear popup menus vs. pie menus

edge of the screen acts like a target with *infinite* size

$T = a + b \log (D/S + 1)$ solves to the minimum of $T=a$



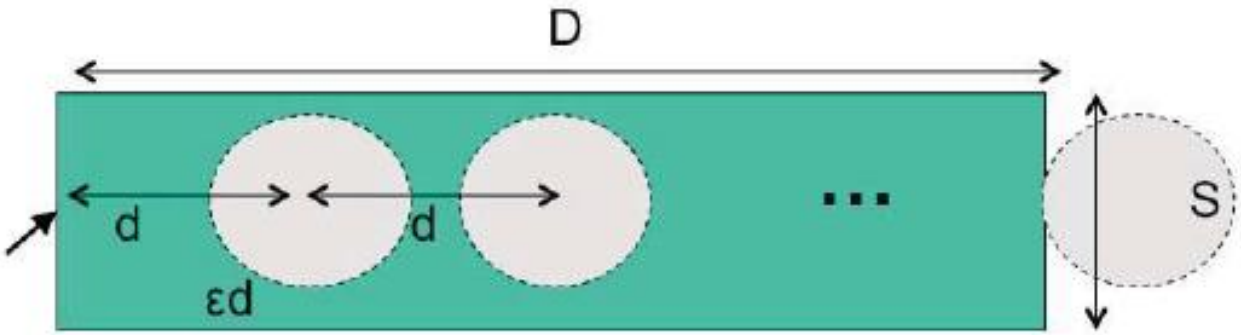
Pie menu efficiency benefits aren't large enough to overcome the external consistency and layout simplicity of linear menus

Pie menus can help you learn gestures, when the same movement of your mouse is used for triggering the pie menu command (note that the Back icon is on the left of the pie menu shown). The combination of pie menus and gestures is called "marking menus"

Steering Tasks

- Time T to move your hand through a tunnel of length D and width S is:

  $$T = a + b\, D/S$$



- Index of difficulty is now *linear*, not logarithmic
  - So steering is much harder than pointing
- Thus cascading submenus are hard to use

Power Law of Practice

- Time T_n to do a task the *n*th time is:

$$T_n = T_1 \, n^{-\alpha}$$

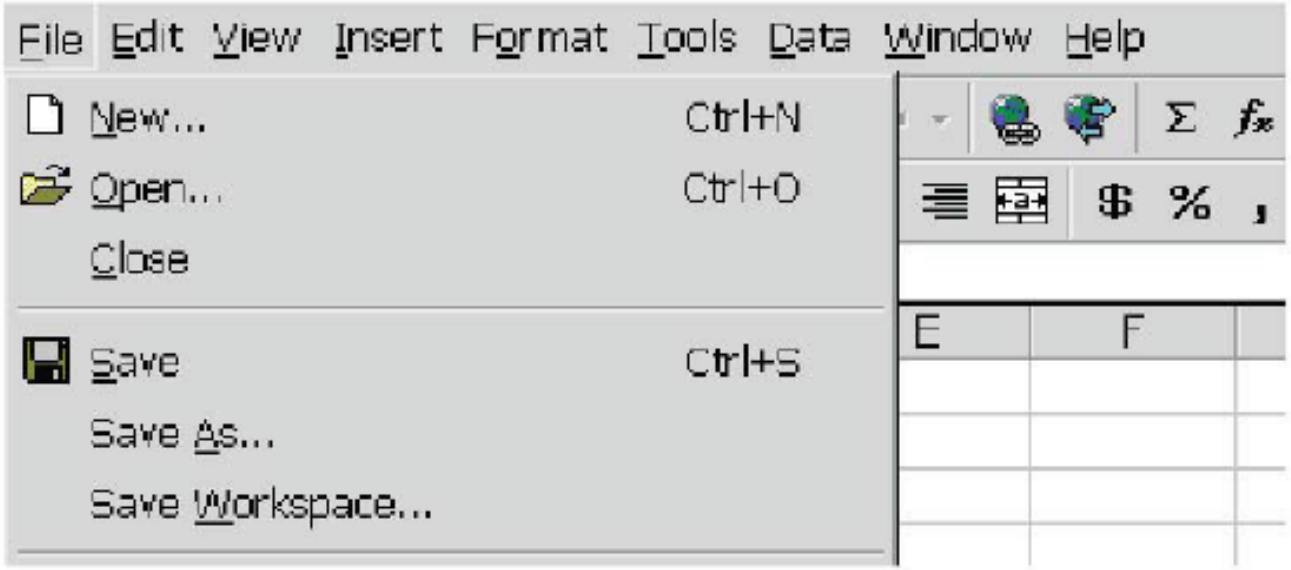
α is typically 0.2-0.6

Improve Mouse Efficiency

- Make frequently-used targets big
  - Use snapping in drawing editors
- Put targets used together near each other
- Use screen corners and screen edges
- Avoid steering tasks

Keyboard Shortcuts

- Keyboard commands
- Menu accelerators

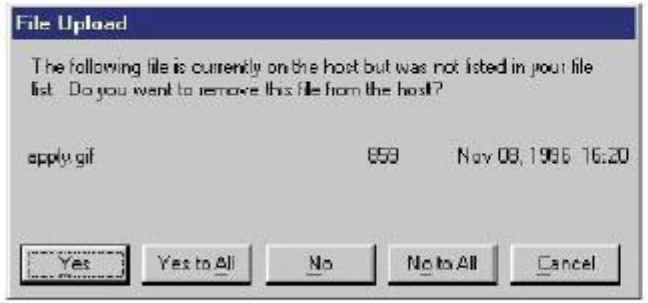


accessibility benefits, since it allows your interface to be used by users who can't see well enough to point a mouse

Command Aggregates

- Styles
- Scripts
- Bookmarks

Aggregating Questions



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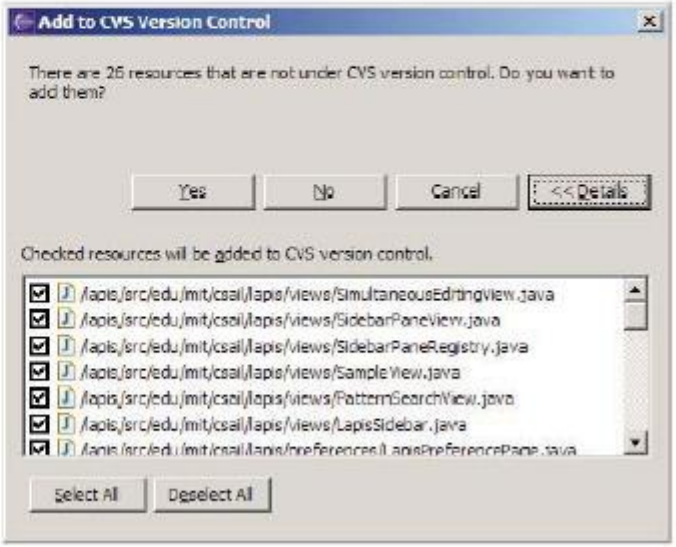

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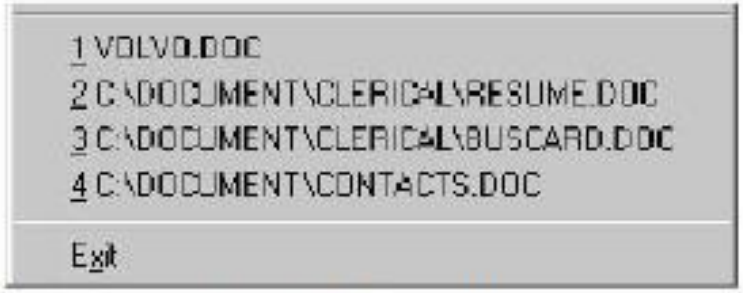
Use Defaults and History

- Use defaults
  - Initially, most likely entry
  - After use, previous entry
- Keep histories



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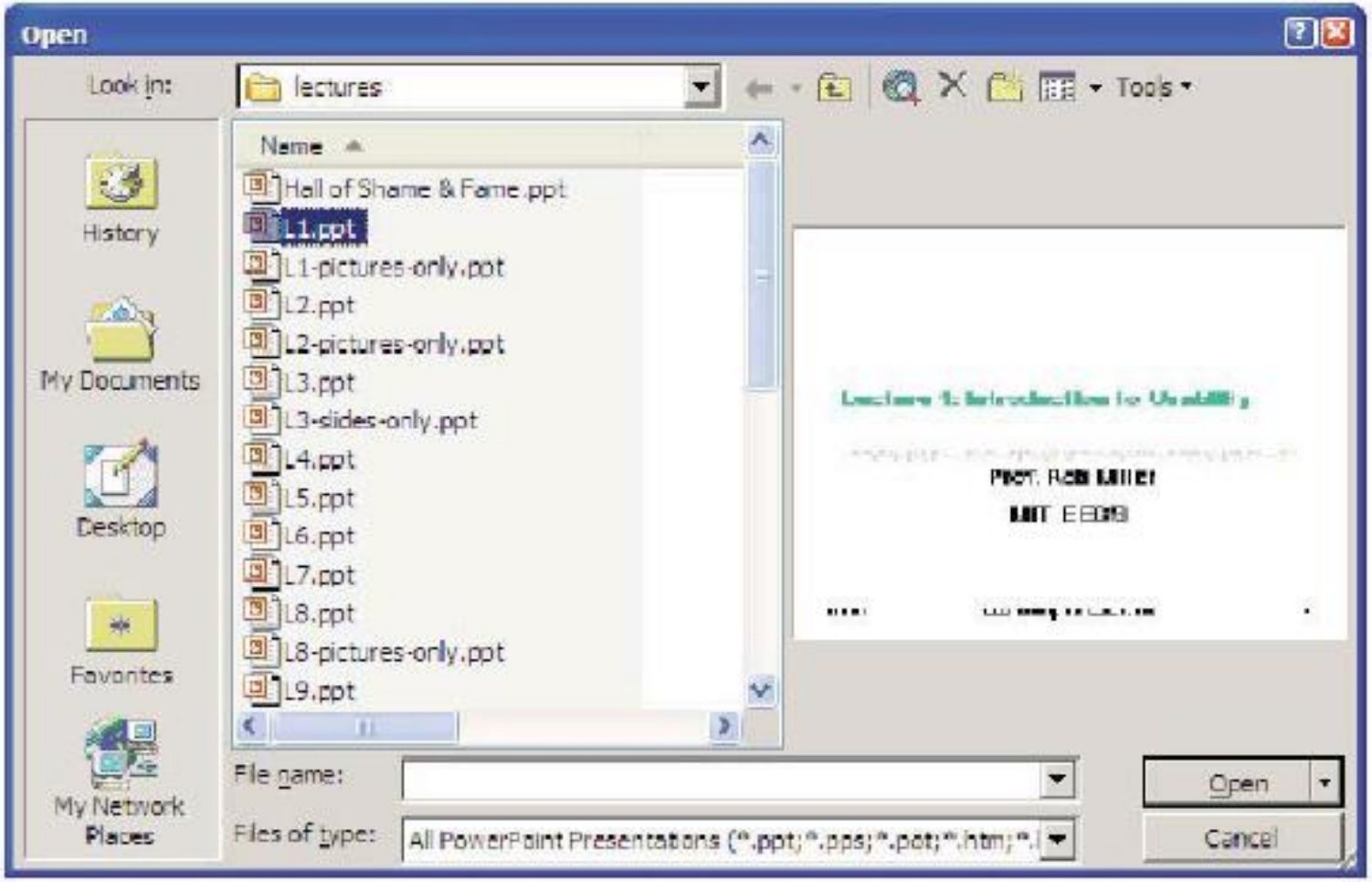
- Offer autocompletion

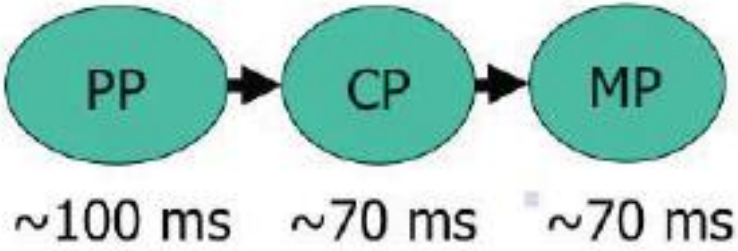
Anticipation



Put related items together. Shortcuts serve **learnability, visibility** and **efficiency**

Predictive Evaluation

- Predictive evaluation uses an engineering model of human cognition to predict usability
  - In this case, we'll predict efficiency
- The engineering model is
  - abstract
  - quantitative
  - approximate
  - estimated from user experiments



Advantages of Predictive Evaluation

- Don't have to build UI prototype
  - Can compare design alternatives with no implementation whatsoever
- Don't have to test real live users
- Theory provides explanations of UI problems
  - So it points to the areas where design can be improved
  - User testing may only reveal problems, not explain them

User testing might show that design A is 25% slower than design B at a particular task, but it won't explain *why*

Predictive evaluation breaks down the user's behavior into little pieces, so that you can actually point at the part of the task that was slower, and see why

Keystroke-Level Model (KLM)

- **K**eystroke
- **B**utton press or release with mouse
- **P**oint with mouse
- **D**raw line with mouse
- **H**ome hands between mouse and keyboard
- **M**entally prepare

KLM Analysis

- Encode a method as a sequence of physical operators (KPHD)
- Use heuristic rules to insert mental operators (M)
- Add up times for each operator to get total time for method

Focus on a particular **method** for doing the task. Suppose the task is deleting a word in a text editor. variety of methods for doing this, e.g.:

(1) click and drag to select the word, then press the Del key

(2) click at the start and shift-click at the end to select the word, then press the Del key;

(3) click at the start, then press the Del key N times

(4) doubleclick the word, then select the Edit/Delete menu command; etc.

Estimated Operator Times

- **K**eystroke determined by typing speed

  0.28 s average typist (40 wpm)

  0.08 s best typist (155 wpm)

  1.20 s worst typist

- **B**utton press or release

  0.1 s highly practiced, no need to acquire button

- **P**ointing determined by Fitts's Law

  $T = a + b \log(d/s + 1) = a + b\ ID$

  0.8 + 0.1 ID [Card 1978]

  0.1 + 0.4 ID [Epps 1986]

  -0.1 + 0.2 ID [MacKenzie 1990, mouse selection]

  0.14 + 0.25 ID [MacKenzie 1990, mouse dragging]

  OR

  T ~ 1.1 s for all pointing tasks

- **D**rawing determined by steering law

- **H**oming estimated by measurement

  0.4 s (between keyboard and mouse)

- **M**ental preparation estimated by measurement

  1.2 s

Heuristic Rules for adding M's

- Basic idea:
  - M before every chunk in the method that must be recalled from long-term memory or that involves a decision
- Before each task or subtask
- Deciding which way to do a task
- Retrieving a chunk from memory
  - Command name
  - File name
  - Parameter value
- Finding something on screen
  - So P is often preceded by M
  - Unless the location is well-known from practice, in which case the visual search is overlapped with the motor action
- Verifying entry or action result
  - e.g. before pressing OK on a dialog

Example: Deleting a Word

- Shift-click selection

  M
  P [start of word]
  BB [click]
  M
  P [end of word]
  K [shift]
  BB [click]
  H [to keyboard]
  M
  K [Del]

- Total: 3M + 2P + 4B + 1K

  = 6.93 sec

- Del key N times

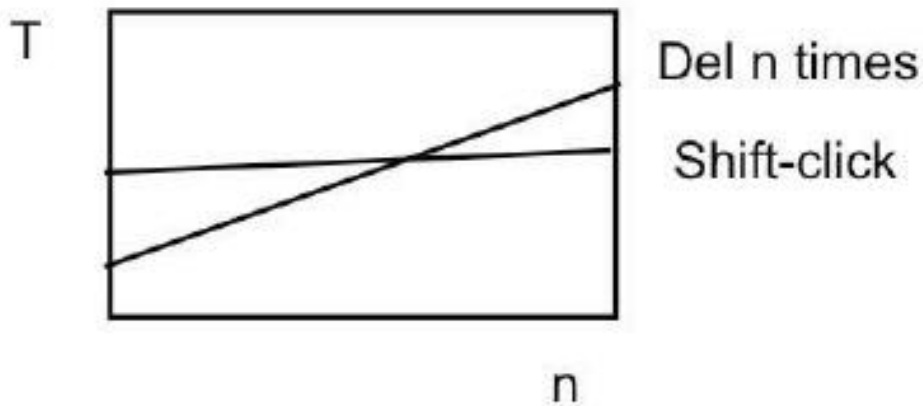
  M
  P [start of word]
  BB [click]
  H
  M
  K [Del]
  x n [length of word]

- Total: 2M+P+2B+H+nK

  = 4.36 + 0.28n sec

Applications of KLM

- Comparing designs & methods
- Parametric analysis



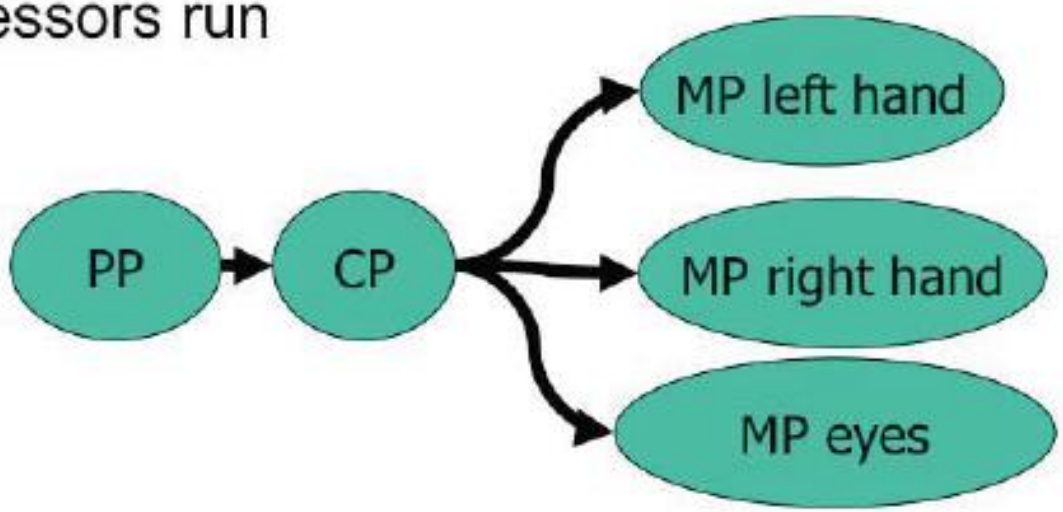
limitation of approximate model: shift-click method isn't really constant with *n* – as the word grows, the distance you have to move the mouse to click at the end of the word grows likewise. keystroke-level approximation assumes that all P operators take constant time. On the other hand, Fitts's Law says that the pointing time would grow at most logarithmically with *n*

Limitations of KLM

- Only expert users doing routine (well-learned) tasks
- Only measures efficiency
  - Not learnability, memorability, errors, etc.
- Ignores
  - errors (methods must be error-free)
  - parallel action (shift-click)
  - mental workload (e.g. attention & WM limits)
  - planning & problem solving (how does user select the method?)
  - fatigue

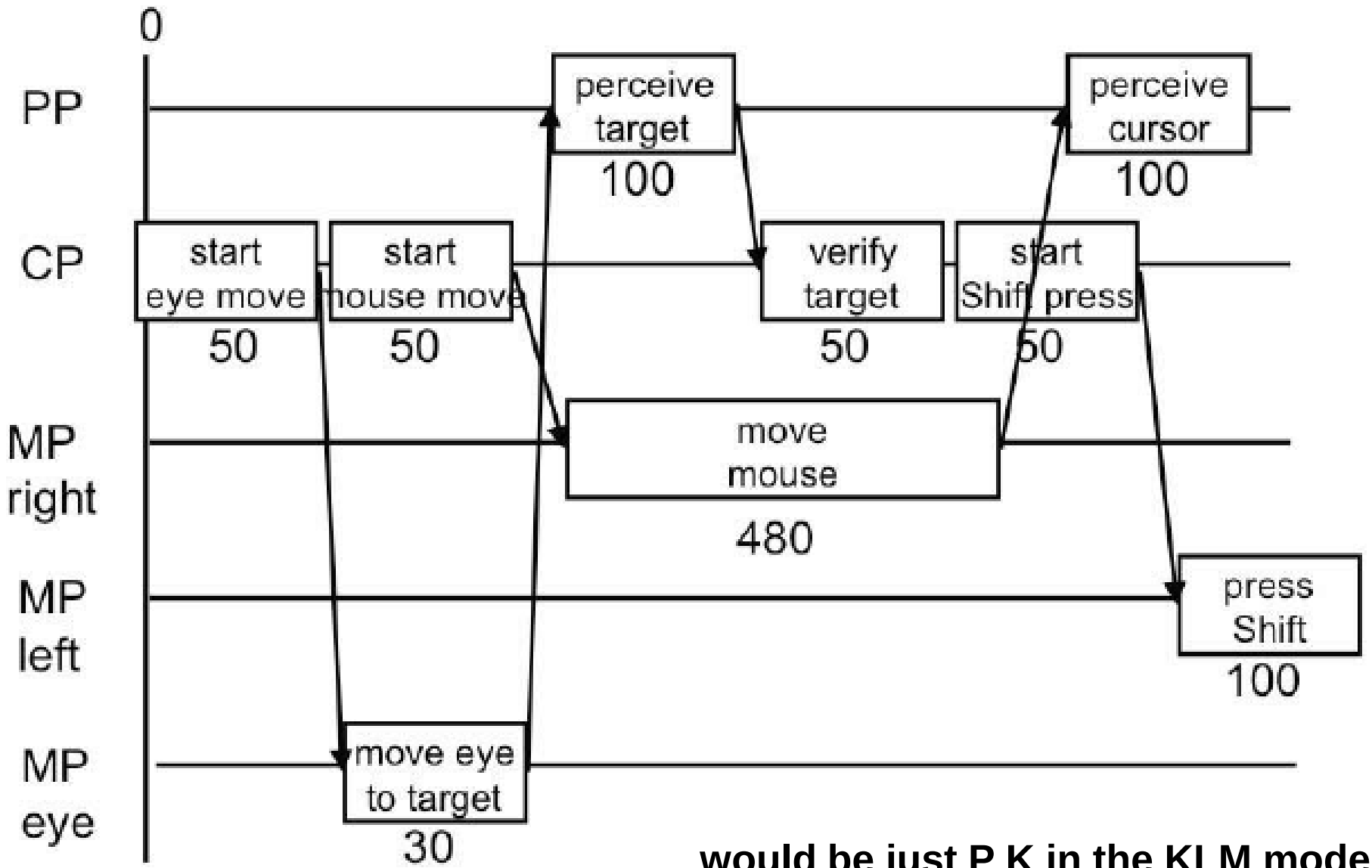
CPM-GOMS

- CPM-GOMS models parallel operations
  - e.g. point & shift-click
- Uses parallel cognitive model
  - each processor is serial
  - different processors run in parallel



Cognitive-Perceptual-Motor

Critical Path Determines Time



would be just P K in the KLM model

Analysis of Phone Operator Workstation

- Phone company considering redesign of a workstation (keyboard + software) for telephone operators (411 service)
  - Reduced keystrokes needed for common tasks
  - Put frequently-used keys closer to user's fingers
- But new design was 4% slower than old design
  = 1 sec/call = $3 million/year
- Keystroke-level model has no explanation
- But CPM-GOMS explained why:
  - Keystrokes removed were not on the critical path
    - Used during slack time, while greeting customer
  - A keystroke was moved from the beginning of call (during slack time) to later (putting it on the critical path)

Summary

- Mouse movement
  - Pointing depends on distance and size
  - Steering is exponentially harder than pointing
- Design principles
  - Shortcuts
  - Defaults, history, and anticipation
- Predictive evaluation uses abstract models to predict usability quantities
  - KLM and CPM-GOMS model efficiency for expert users